Software Requirement Specification (SRS)

IntelliPlant AI

AI-Powered Industrial Knowledge Intelligence Platform

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1. Introduction

IntelliPlant AI is an AI-powered Industrial Knowledge Intelligence Platform designed to ingest structured and unstructured industrial documents and convert them into a searchable and actionable knowledge system.

The platform combines:

- OCR
- Knowledge Graphs
- Hybrid RAG
- Multi-Agent AI
- Root Cause Analysis
- Compliance Intelligence
- Lessons Learned Engine

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2. Purpose

The purpose of this system is to:

- Eliminate information silos.
- Preserve tribal knowledge.
- Improve maintenance decision making.
- Reduce downtime.
- Enhance regulatory compliance.
- Provide explainable AI-powered assistance.

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3. Scope

The platform shall:

Support

- PDF
- DOCX
- XLSX
- CSV
- Images
- Email archives

The platform shall provide:

- Semantic search
- Conversational AI
- Knowledge graph
- Root cause analysis
- Compliance intelligence
- Lessons learned engine

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4. Definitions

Term	Description
RAG	Retrieval Augmented Generation
OCR	Optical Character Recognition
RCA	Root Cause Analysis
KG	Knowledge Graph
LLM	Large Language Model
SOP	Standard Operating Procedure
CAPA	Corrective and Preventive Action
OISD	Oil Industry Safety Directorate

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5. Overall Description

The system consists of:

Frontend

NextJS Dashboard

Backend

FastAPI

AI Layer

LangGraph Multi-Agent System

Vector Database

Qdrant

Graph Database

Neo4j

Metadata Database

PostgreSQL

OCR Layer

Docling + PaddleOCR

LLM Layer

Gemini/OpenAI

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6. Product Functions

The system shall provide:

Document Ingestion

Upload and process documents.

OCR

Extract text from images and scanned documents.

Entity Extraction

Identify:

- Equipment
- Failure modes
- Dates
- Personnel
- Regulations

Knowledge Graph Generation

Generate relationships between entities.

Conversational Search

Natural language query support.

Root Cause Analysis

Failure investigation assistance.

Compliance Intelligence

Regulatory gap detection.

Lessons Learned Engine

Pattern analysis across incidents.

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7. User Classes

Administrator

Manage users and system configuration.

Maintenance Engineer

Analyze equipment failures.

Reliability Engineer

Perform RCA.

Quality Engineer

Perform audits and compliance checks.

Plant Manager

Monitor operational intelligence.

Field Technician

Mobile knowledge access.

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8. Assumptions

Documents are available digitally.

Internet connectivity exists.

Users possess domain knowledge.

LLM services are available.

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9. Constraints

Large document sizes.

Complex P&ID diagrams.

Domain-specific terminology.

Limited training datasets.

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10. Functional Requirements

FR-001 Document Upload

Priority: High

Description:

The system shall support document upload.

Supported formats:

- PDF
- DOCX
- XLSX
- CSV
- JPG
- PNG

FR-002 OCR Processing

Priority: High

The system shall extract text from:

- Images
- Scanned PDFs
- P&ID drawings

FR-003 Metadata Extraction

Priority: High

Extract:

- Equipment IDs
- Dates
- Operators
- Process values
- Failure information

FR-004 Chunk Generation

Priority: High

Documents shall be split into semantic chunks.

FR-005 Embedding Generation

Priority: High

Generate embeddings using BGE models.

FR-006 Vector Storage

Priority: High

Store embeddings inside Qdrant.

FR-007 Knowledge Graph Creation

Priority: High

Create relationships among:

- Equipment
- Incidents
- Manuals
- Regulations

- Personnel

FR-008 Hybrid Search

Priority: High

Support:

- Vector search
- BM25 search
- Graph search

FR-009 Conversational AI

Priority: High

Provide answers with:

- Source citations
- Confidence score
- Related entities

FR-010 Root Cause Analysis

Priority: High

Recommend probable causes based on:

- Similar incidents
- Manuals
- Historical failures

FR-011 Compliance Intelligence

Priority: High

Evaluate:

- Factory Act
- OISD

- ISO standards
- PESO regulations

FR-012 Lessons Learned Engine

Priority: Medium

Analyze:

- Incidents
- Near misses
- Audit reports

Generate recommendations.

FR-013 Multi-Agent Workflow

Priority: High

Agents:

- Supervisor Agent
- Expert Copilot Agent
- RCA Agent
- Compliance Agent
- Lessons Agent

FR-014 Authentication

Priority: High

Support:

- Email login
- JWT
- RBAC

FR-015 Audit Logs

Priority: Medium

Track:

- User actions
- Queries
- Uploads

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11. Non-Functional Requirements

NFR-001 Performance

Average response time:

<5 seconds

NFR-002 Availability

99% uptime

NFR-003 Scalability

Support:

1000+ users

NFR-004 Reliability

Fault tolerant architecture.

NFR-005 Maintainability

Modular microservices.

NFR-006 Security

JWT authentication

Encrypted communication

Role-based access

NFR-007 Usability

Minimal learning curve.

NFR-008 Explainability

Every answer shall include citations.

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12. External Interface Requirements

User Interface

Web dashboard

Mobile responsive UI

APIs

REST APIs

Future GraphQL support

Database Interfaces

PostgreSQL

Neo4j

Qdrant

Redis

External Services

Gemini API

OpenAI API

PaddleOCR

Docling

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13. Data Requirements

Entities:

Equipment

Incident

Manual

Inspection

Operator

Regulation

Failure

Work Order

Relationships:

CAUSES

CONNECTED_TO

FAILED_IN

REQUIRES

DESCRIBED_BY

SIMILAR_TO

14. Security Requirements

Authentication:

JWT

Authorization:

RBAC

Encryption:

HTTPS

Password Hashing:

bcrypt

Audit Logging:

Enabled

Input Validation:

Pydantic

Rate Limiting:

Enabled

15. Performance Requirements

Metric	Target
Search latency	<2 sec
Chat response	<5 sec
Upload processing	<30 sec
OCR speed	<20 sec
Concurrent users	1000

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16. Reliability Requirements

Automatic retry mechanism.

Error handling.

Graceful degradation.

Backup strategy.

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17. Availability Requirements

Target uptime:

99%

Recovery Time Objective:

15 minutes

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18. Scalability Requirements

Horizontal scaling.

Containerized deployment.

Distributed vector search.

Caching with Redis.

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19. Use Cases

UC-01 Upload Document

Actor:

Engineer

Precondition:

Authenticated user

Flow:

Upload file

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OCR

↓

Chunking

↓

Embedding

↓

Vector DB

↓

Knowledge Graph

Postcondition:

Document searchable.

UC-02 Ask Question

Actor:

Engineer

Flow:

Ask query

↓

Hybrid Retrieval

↓

Multi-Agent Routing

↓

LLM

↓

Answer with citations

UC-03 Root Cause Analysis

Actor:

Reliability Engineer

Flow:

Input equipment failure

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Retrieve similar failures

↓

Analyze manuals

↓

Generate RCA

↓

Recommend actions

UC-04 Compliance Check

Actor:

Quality Engineer

Flow:

Load documents

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Compare regulations

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Identify gaps

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Generate audit evidence

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20. Acceptance Criteria

System must:

✓ Process PDFs

✓ Support OCR

✓ Generate embeddings

✓ Build knowledge graph

✓ Answer with citations

✓ Support multi-agent workflow

✓ Provide confidence score

✓ Detect compliance gaps

✓ Perform RCA

✓ Generate lessons learned

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21. Requirement Traceability Matrix

Requirement	Module
FR-001	Ingestion Service
FR-002	OCR Service
FR-003	Entity Extraction
FR-004	Chunking Engine
FR-005	Embedding Engine
FR-006	Qdrant
FR-007	Neo4j
FR-008	Hybrid Search
FR-009	Chat Engine
FR-010	RCA Agent
FR-011	Compliance Agent
FR-012	Lessons Agent
FR-013	LangGraph Supervisor
FR-014	Auth Service
FR-015	Audit Service

Conclusion

IntelliPlant AI provides an enterprise-grade Industrial Knowledge Intelligence Platform capable of transforming heterogeneous documents into a unified operational brain through OCR, GraphRAG, multi-agent AI, semantic search, and explainable intelligence.